IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

SIGNALS AND SYSTEMS

Friday, 3 May 2:30 pm

Time allowed: 2 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : P. Dragotti

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Signals and Systems

Formulae

Standard Laplace Transforms

Signal	Transform	ROC
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$\delta(t-\tau),\ \tau \geq 0$	$e^{-s\tau}$	$\forall s$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -a$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -a$
$\cos(w_1 t)u(t)$	$\frac{s}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$\sin(w_1 t)u(t)$	$\frac{w_1}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$e^{-at}\cos(w_1 t)u(t)$	$\frac{s+a}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$
$e^{-at}\sin(w_1 t)u(t)$	$\frac{w_1}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$

Standard z-transform pair

$$x[n-n_0] \overset{\mathrm{z}}{\longleftrightarrow} z^{-n_0}X(z) \quad \text{ROC: } R_x \text{ but may exclude } z=0 \text{ or } \infty$$

1. *This question caries 40% of the marks.*

a) Explain the meaning and the key properties of

i) a quantized signal, [2]

ii) a continuous-time signal, [2]

iii) a discrete-time signal. [2]

b) Give an example of a random signal encountered in the physical world. [2]

c) For each of the following continuous-time signals, state whether they are causal or noncausal, periodic or nonperiodic, and even, odd or non-symmetric. Give justifications for your statements. If periodic, state the period.

i) $x(t) = \sin^2(t/2 + \pi)$ [2]

ii) $x(t) = \int_{-\infty}^{t} \delta(\tau) d\tau$ [2]
where the function $\delta(t)$ is defined as usual.

iii) $x(t) = \begin{cases} 3e^{-0.5t} & t \geq 0 \\ 0 & \text{otherwise.} \end{cases}$ [2]

d) Consider the signal

$$x(t) = \begin{cases} 1-t & 0 \leq t \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

For each of the following signals $y(t)$, find an expression for $y(t)$ not involving x. Describe the sequence of steps that must be taken to form $y(t)$ from $x(t)$. Show the sequence of steps in illustrative sketches.

i) $y(t) = x(0.25t + 1)$ [3]

ii) $y(t) = x(-2t + 2)$ [3]

e) A real-valued signal $x(t)$ has Fourier transform $X(j\omega)$. A second signal $q(t)$ has Fourier transform $Q(j\omega)$. These signals are related as

$$q(t) = x(-t).$$

i) Derive an expression for $Q(j\omega)$ in terms of $X(j\omega)$, and show your working. [3]

ii) State how the magnitude spectra of $X(j\omega)$ and $Q(j\omega)$ are related. [1]

f) The discrete-time signal $y[n]$ is generated from a discrete-time signal $x[n]$, so that

$$y[n] = x[n] + ay[n-2]$$

where a is a scalar constant.

Transform this operation into the z-transform domain so as to obtain the corresponding expression for $Y(z)$. Write your expression so that any powers of z are only positive powers of z. [4]

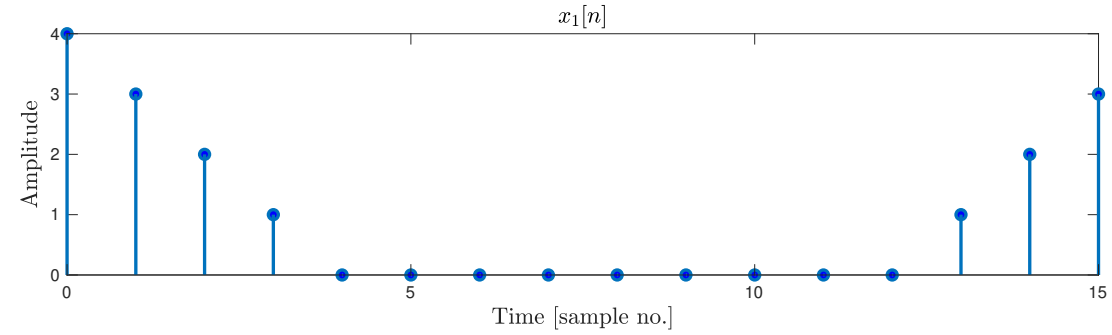


Figure 1.1

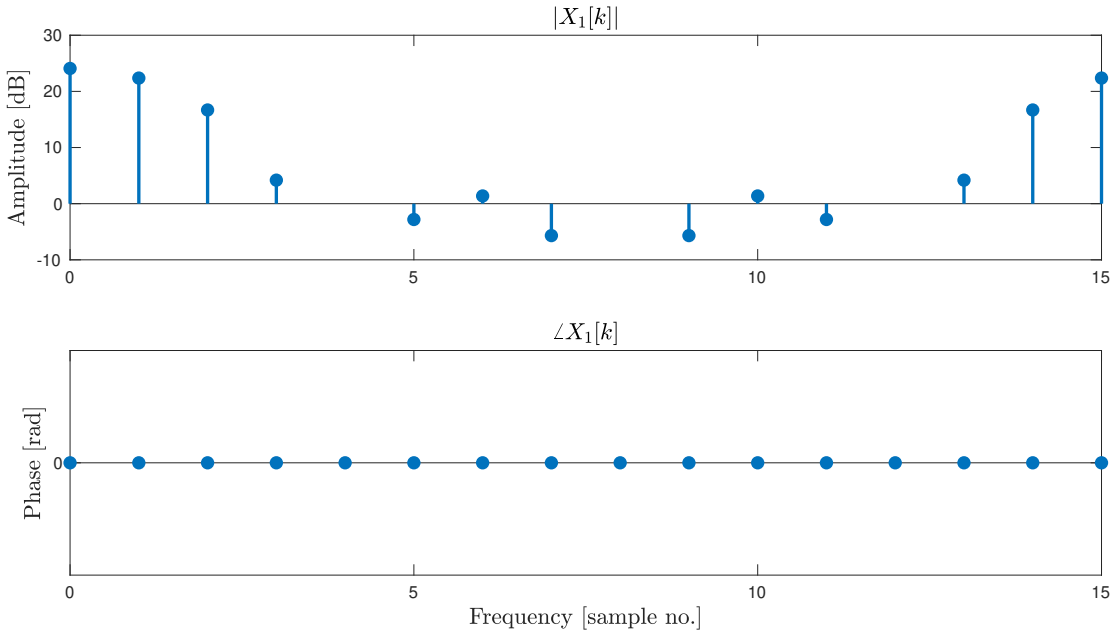


Figure 1.2

g) Consider the signal $x_1[n]$ shown in Fig. 1.1. The magnitude and phase spectra are computed using a 16-point DFT and are shown in Fig. 1.2. The sampling frequency used in this question is 8 kHz.

i) What is the frequency resolution, Δf, of the 16-point DFT? [1]

ii) Is the frequency domain signal representation, $X_1[k]$, purely real, purely imaginary, or complex? Justify your answer using properties of the time-domain signal. [2]

iii) Now consider the delayed signal $x_2[n]$ plotted in Fig. 1.3. The 16-point DFT of $x_2[n]$ is taken to obtain $X_2[k]$. Describe similarities and differences between the spectra $X_2[k]$ and $X_1[k]$, stating any properties of the Fourier transform supporting your argument. [6]

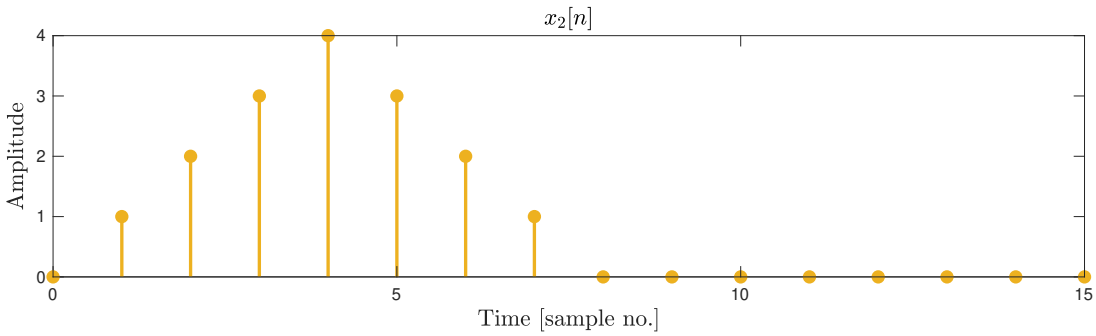


Figure 1.3

iv) Now consider the signal $x_3[n]$ plotted in Fig. 1.4. The 16-point DFT of $x_3[n]$ is taken to obtain $X_3[k]$. Describe the phase of $X_3[k]$ and give a mathematical expression for $\angle X_3[k]$. [3]

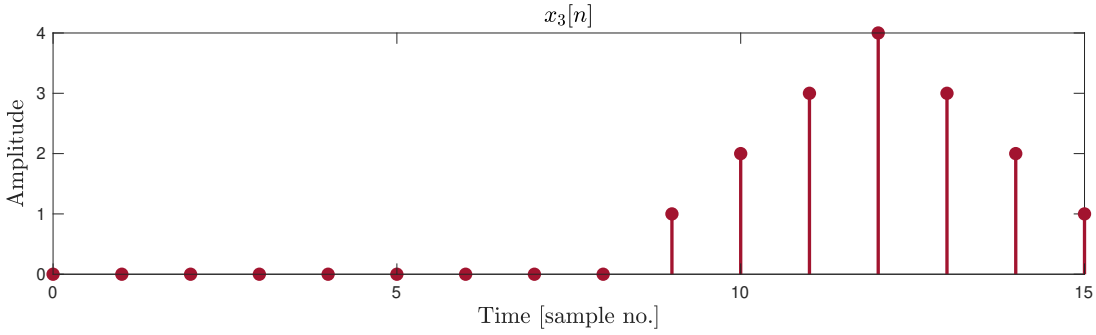


Figure 1.4

2. *This question caries 30% of the marks.*

a) Consider $y(t)$ to be the product of two continuous-time signals such that

$$y(t) = x(t)p(t).$$

Let $Y(\omega)$, $X(\omega)$ and $P(\omega)$ denote the Fourier transforms of $y(t)$, $x(t)$ and $p(t)$ respectively.

i) State the Multiplication Property of the Fourier transform and hence write down an expression for $Y(\omega)$ in terms of $X(\omega)$ and $P(\omega)$. [4]

ii) Let $p(t) = \cos\omega_0 t$. Write down the corresponding expression for $P(\omega)$. Hence, express $Y(\omega)$ as a function of only $X(\omega)$.[4]

b) Consider the signal

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}.$$

i) Deduce the key property of $x(t)$, the meaning of ω_0 and the units of ω_0. [3]

ii) Write a corresponding expression for $x^*(t)$ where the superscript $*$ indicates the complex conjugate. [3]

iii) Show that $x(t)$ is real if $a_k^* = a_{-k}$. [2]

iv) Now consider a particular $x(t)$ with:
$a_{-3} = \frac{1}{4}$, $a_{-2} = \frac{1}{3}$, $a_{-1} = \frac{1}{2}$, $a_0 = 1$, $a_1 = \frac{1}{2}$, $a_2 = \frac{1}{3}$, $a_3 = \frac{1}{4}$,
and $a_k = 0$ for $|k| > 3$.

Write $x(t)$ as a sum of sine and cosine functions and hence draw an indicative sketch of $x(t)$. [6]

c) A particular LTI system with input signal $x(t)$ and output signal $y(t)$ is described by

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 2x(t).$$

i) Show that the frequency response of this system is

$$H(j\omega) = \frac{j\omega + 2}{(j\omega)^2 + 4j\omega + 3}.$$

Clearly explain each step of your solution. [2]

ii) For an input signal $x(t) = e^{-t}u(t)$, write down its spectrum $X(j\omega)$. [1]

iii) Write down a frequency domain expression for the output spectrum, $Y(j\omega)$, when the input signal is as defined in part ii). [2]

iv) Find the time domain expression for the output signal, $y(t)$, using partial fraction expansion to determine the inverse Fourier transform of $Y(j\omega)$. [3]

3. *This question caries 30% of the marks.*

a) The discrete Fourier transform for a signal $x[n]$ can be written

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} \quad k = 0, 1, \ldots, N-1.$$

i) State the meaning of the variables n, N, and k. [3]

ii) Explain the meaning of $X(3)$ for the case $N = 8$. Is $X(3)$ a real-valued or complex-valued quantity? [3]

iii) Given that $x[n] = [1, -1, 1, -1, 1, -1, 1, -1]$, determine $X(3)$ and $X(4)$. [4]

b) i) Consider the causal discrete-time system illustrated in Fig. 3.1.

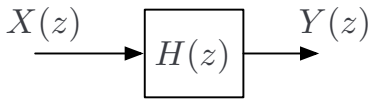


Figure 3.1

Given that

$$H(z) = \frac{z}{z^2 - (0.1 - a)z - 0.1a},$$

find the range of values of a for which the system is stable. [4]

ii) Using the standard z-transform pair given in the formulae list above, or otherwise, find the difference equation for $y[n]$ for the system of part i). [4]

iii) Now consider the causal discrete-time system illustrated in Fig. 3.2.

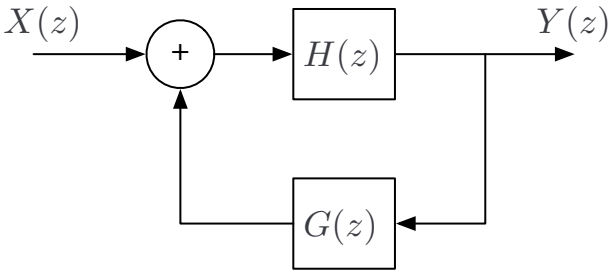


Figure 3.2

Given that

$$H(z) = \frac{z}{z - 0.1} \quad \text{and} \quad G(z) = kz^{-1},$$

find the range of values of k for which the system is stable. [4]

c) A causal, discrete-time, 2nd-order FIR filter $P(z)$ operates with a sampling frequency of 8 kHz. The filter is designed to attenuate an interfering signal with a frequency of 2 kHz.

i) Deduce and sketch a z-plane plot of a suitable filter's system function $P(z)$. [4]

ii) Determine and write down the corresponding z-domain expression for $P(z)$ for the filter you deduced in part i). [4]